

The diaphragm is a domed musculo-fibrous sheet that separates the thoracic and abdominopelvic cavities (Fig. 55.1). Its mainly convex superior surface faces the thorax, and its concave inferior surface is directed towards the abdomen. The positions of the domes or cupulae of the diaphragm are extremely variable; they are influenced by body build (often higher in the short and obese compared to the tall and asthenic), position and ventilatory phase. Conditions resulting in lung over-inflation, e.g. emphysema, cause marked diaphragmatic depression. Usually, after forced expiration, the right cupula is level anteriorly with the fourth costal cartilage and the right nipple, whereas the left cupula lies approximately one rib lower. On full inspiration, the cupula will descend by as much as 10 cm, and on a plain chest radiograph the right dome coincides with the anterior end of the sixth rib. The diaphragm lies more superiorly in the supine compared to the erect position, and the dependent half of the diaphragm will be considerably higher than the uppermost one in the decubitus position.

ATTACHMENTS AND COMPONENTS

The muscle fibres of the diaphragm arise from the highly oblique circumference of the thoracic outlet; the attachments are low posterolaterally and high anteriorly (Fig. 55.2). Although it is a continuous sheet, the muscle can be considered in three parts: sternal, costal and lumbar. The sternal part arises by two fleshy slips from the posterior aspect of the xiphoid process. The attachment varies; it may be a broad, continuous band but is not always present (Loukas et al 2015a). On each

side, the costal part arises from the internal surfaces of the lower six costal cartilages and their adjoining ribs, interdigitating with transversus abdominis (see Fig. 53.15).

The lumbar part arises from two aponeurotic arches – the medial and lateral arcuate ligaments (sometimes termed lumbocostal arches) – and asymmetrically from the upper lumbar vertebral bodies via two pillars or crura. The lateral arcuate ligament covers quadratus lumborum, arching across the upper part of that muscle, and attaching medially to the anterior aspect of the transverse process of the first lumbar vertebra, and laterally to the inferior margin of the midpoint of the twelfth rib. The medial arcuate ligament covers the upper part of psoas major. Medially, it is continuous with the lateral tendinous margin of the corresponding crus, and is thus attached to the side of the body of the first or second lumbar vertebra. Laterally, it is fixed to the anterior aspect of the transverse process of the first lumbar vertebra.

The crura are tendinous at their attachments and blend with the anterior longitudinal ligament. The broader and longer right crus arises from the anterolateral surfaces of the bodies and intervertebral discs of the upper three lumbar vertebrae. The left crus arises from the corresponding parts of the upper two lumbar vertebrae. The medial tendinous margins of the crura meet in the midline to form an often poorly defined arch, the median arcuate ligament, which crosses anterior to the aorta at the level of the intervertebral T12/L1 disc. The crura vary greatly in length, thickness and ratio of muscle to tendon (Loukas et al 2015a).

From these circumferential attachments, the muscle fibres of the diaphragm converge into a central tendon. Fibres from the xiphoid

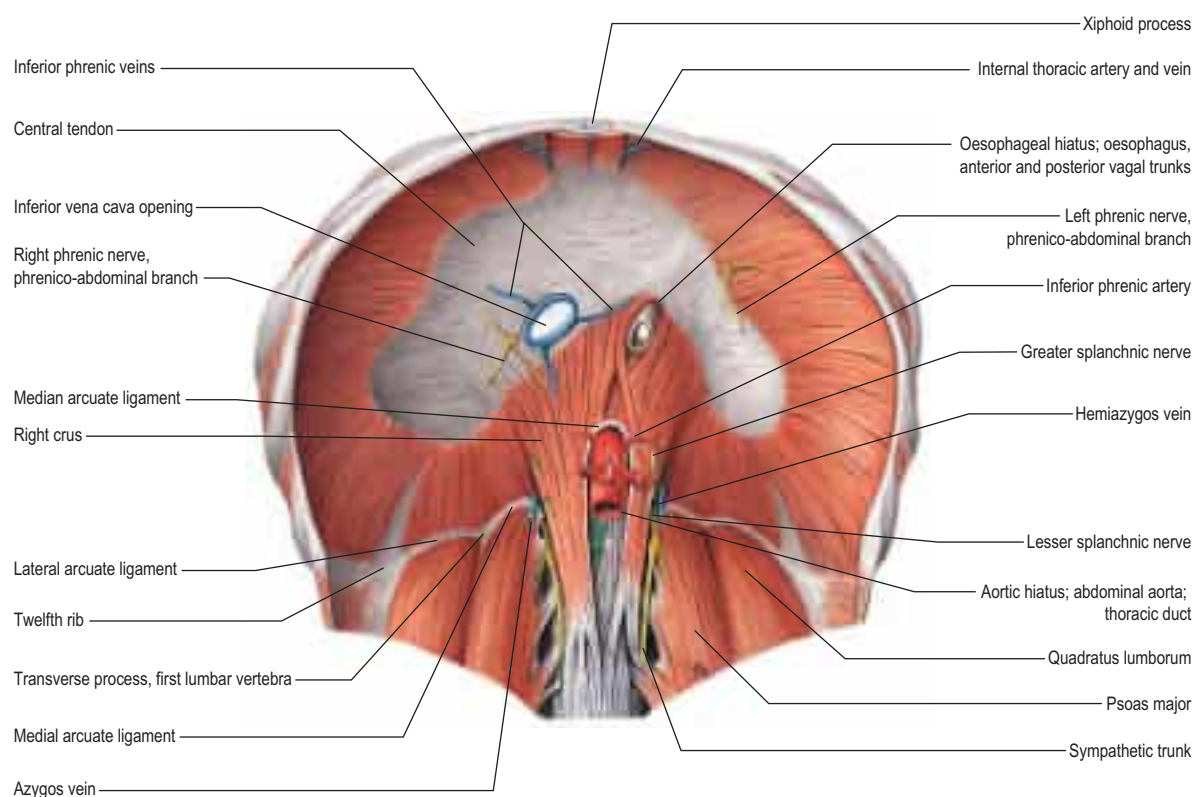


Fig. 55.1 The abdominal aspect of the diaphragm. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)



Fig. 55.2 A coronal CT scan depicting the diaphragmatic domes (arrows) superior to the liver and spleen. Notice the attachment of the diaphragm at the lumbar vertebrae. (Courtesy of Professor Loukas.)

process are short, run almost horizontally and are occasionally aponeurotic. Fibres originating from the medial and lateral arcuate ligaments, and particularly the osteochondral surfaces of the ribs, are much longer. They arise almost vertically at first and then curve towards their central attachment. Fibres from the crura diverge, the most lateral becoming even more lateral as they ascend to the central tendon. Medial fibres of the right crus embrace the oesophagus where it passes through the diaphragm, the more superficial fibres ascend on the left, and deeper fibres cover the right margin. Sometimes, a fleshy fasciculus from the medial side of the left crus crosses the aorta and runs obliquely through the fibres of the right crus towards the caval opening; this fasciculus does not continue upwards around the oesophageal hiatus on the right side.

The central tendon of the diaphragm is a thin but strong aponeurosis of closely interwoven fibres situated near the centre of the muscle, but closer to the anterior wall of the thorax, so that the posterior muscular fibres are longer (Fig. 55.3). In the centre, it lies immediately inferior to the pericardium, with which it is partially blended. It is trifoliate in shape. The middle folium, which projects anteriorly, has the form of an equilateral triangle with the apex directed towards the xiphoid process. The right and left folia are tongue-shaped and curve posterolaterally, the left being a little narrower.

The central area of the tendon consists of four well-marked diagonal bands that fan out from a thick central node where compressed tendinous strands decussate anterior to the oesophagus and to the left of the inferior vena cava. Although always present, their morphology varies; in some individuals, the right leaflet of the central tendon is smaller than the left or the anterior leaflets. The ratio between muscle and tendon varies disproportionately, such that larger diaphragms may have a small central tendon, while smaller diaphragms may have only a small amount of muscle contributing to their overall size (Loukas et al 2015a).

PROFILE AND RELATIONS

The superior surface of the diaphragm is related to three serous membranes. On each side, the pleura separates it from the base of the corresponding lung, and the pericardium is interposed between the middle folium of the central tendon and the heart. The middle folium, the 'cardiac plateau', is almost flat and extends more to the left than the right. In anteroposterior view, the superior profile of the diaphragm rises on either side of the cardiac plateau to a smooth convex dome or cupula. The right cupula is higher and slightly broader than the left. Most of the inferior surface is covered by peritoneum. The right side is moulded over the convex surface of the right lobe of the liver, right kidney and suprarenal gland. The left side conforms to the left lobe of the liver, gastric fundus, spleen, left kidney and suprarenal gland. In view of these differences in the profile and anatomical relationships of the right and left sides of the diaphragm, the side should always be

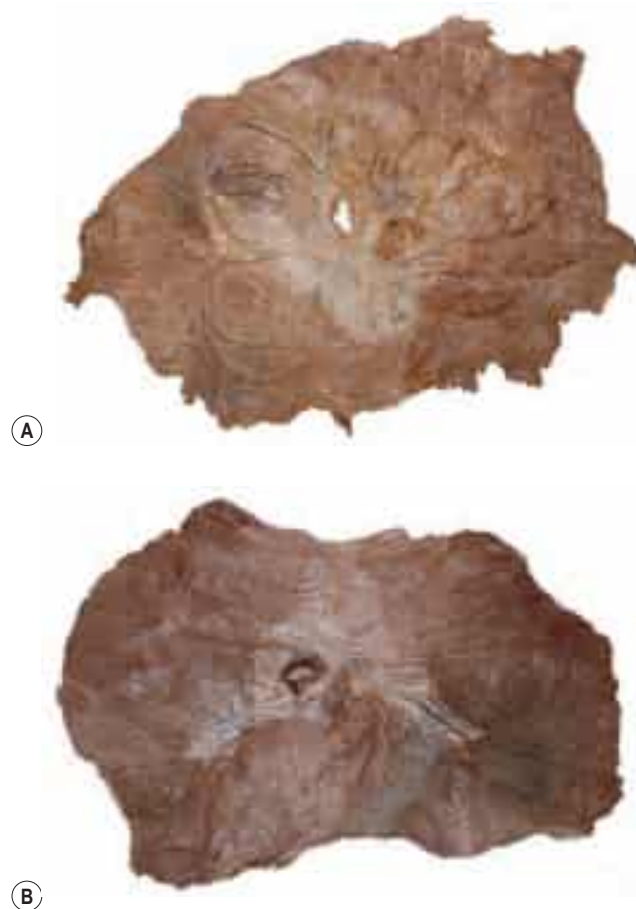


Fig. 55.3 Two diaphragms removed from the thoracic cavity to show the central tendon and muscular components. Note the different proportions of the central tendon and the muscular portions between A and B. (With permission from Loukas M, du Plessis M, Shah S, Tubbs RS, Raffi E, Polo L, Shoja MM. The clinical anatomy of the musculotendinous part of the diaphragm. Surgical and Radiologic Anatomy, in press, 2015a.)

specified in clinical descriptions. The right hemidiaphragm is found at the anterior end of the sixth rib on a properly inspired posteroanterior chest radiograph, the left hemidiaphragm 1.5–2.5 cm lower (see Fig. 56.16). Unilateral paralysis may be seen as a raised hemidiaphragm on a chest radiograph, but this sign should not be relied on uncritically in clinical practice.

Ultrasound imaging The posterolateral aspects of the diaphragm may be visualized using ultrasound. It is typically identified by its curved morphology, deep location and specific echotexture. It can be visualized as a thin, muscular hypoechoic line wedged between two hyperechoic layers consisting of peritoneum and pleura. During inspiration, the hypoechoic line thickens as the muscle contracts, making it more visible. This change can be used diagnostically to identify an atrophic diaphragm, when the size of the hypoechoic line does not change. Ultrasound investigation of the diaphragm is best done with the patient in the supine position, limiting veiling by other organs, as well as the risk of misdiagnosis attributable to underlying pulmonary pathologies. The left side poses more of a challenge in visualization (Sarwal et al 2013).

APERTURES

A number of structures pass between the thorax and abdomen via apertures in the diaphragm. There are three large openings, for the aorta, oesophagus and inferior vena cava, and a number of smaller ones (see Fig. 55.1).

The aortic hiatus is the most posteroinferior of the large openings, and is found at the level of the lower border of the twelfth thoracic vertebra and the adjacent intervertebral disc, slightly to the left of the midline. It is actually an osseo-aponeurotic opening defined by the diaphragmatic crura laterally, the vertebral column posteriorly and the median arcuate ligament anteriorly; the aorta therefore passes posterior to the diaphragmatic musculature and is not affected by its contraction

(Nason et al 2012). Occasionally, some tendinous fibres from the medial parts of the crura also pass posterior to the aorta, converting the osseo-aponeurotic opening into a fibrous ring. The aortic hiatus transmits the aorta, thoracic duct, lymphatic trunks from the lower posterior thoracic wall and, sometimes, the azygos and hemiazygos veins.

The oesophageal hiatus is located at the level of the tenth thoracic vertebra, anterosuperior to, and a little to the left of, the aortic hiatus. It transmits the oesophagus, vagal trunks and gastric nerves, oesophageal branches of the left gastric vessels and some lymphatic vessels. The elliptical opening has a slightly oblique long axis, and is bounded by muscle fibres that originate from the medial part of the right crus and cross the midline, forming a 'chimney' approximately 2.5 cm long, that accommodates the terminal part of the oesophagus. The outermost fibres run in a craniocaudal direction, and the innermost fibres are arranged circumferentially. There is no direct continuity between the oesophageal wall and the muscle around the oesophageal hiatus. The fascia on the inferior surface of the diaphragm, rich in elastic fibres, is continuous with the transversalis fascia and extends superiorly into the opening as a flattened cone to blend with the oesophageal wall 2–3 cm superior to the gastro-oesophageal (squamocolumnar) junction. Some of its elastic fibres penetrate to the submucosa of the oesophagus. This peri-oesophageal areolar tissue, the phreno-oesophageal ligament, connects the oesophagus flexibly to the diaphragm, thus permitting freedom of movement during swallowing and ventilation while synchronously limiting superior oesophageal displacement.

The caval opening, the most superior of the three large openings, lies at about the level of the intervertebral disc between the eighth and ninth thoracic vertebrae. It is quadrilateral, with aponeurotic margins, located at the junction of the right leaf with the central area of the tendon. It is traversed by the inferior vena cava, which adheres to the margin of the opening, and by some branches of the right phrenic nerve. During inspiration, dilation of the caval hiatus, combined with increased intra-abdominal pressure, increases cardiac venous return (Nason et al 2012).

There are two lesser apertures within each crus; one transmits the greater, and the other, the lesser, splanchnic nerve. The ganglionated sympathetic trunks usually enter the abdominal cavity posterior to the diaphragm, deep to the medial arcuate ligaments. Openings for minute veins frequently occur in the central tendon.

On each side of the diaphragm, there are small areas where the muscle fibres are replaced by areolar tissue. One, between the sternal and costal parts, contains the superior epigastric branch of the internal thoracic artery and some lymph vessels from the abdominal wall and convex surface of the liver. The other, between the costal part and the fibres that spring from the lateral arcuate ligament, is less constant; when it is present, the posterolateral surface of the kidney is separated from the pleura only by areolar tissue. Additional, smaller fascial spaces may also be found throughout the lateral aspects of the diaphragmatic domes without evidence of vascular penetration. Although these spaces are found on both left and right sides, they appear to be more numerous on the left (Loukas et al 2015a).

Oesophageal reflux Reflux of gastric contents into the oesophagus, with risk of pulmonary aspiration, is normally prevented by a physiological antireflux barrier located at the gastro-oesophageal junction. The major components of this barrier consist of the specialized smooth muscle of the inferior oesophageal wall and the encircling crural fibres. The diaphragm acts as a 'physiological' sphincter to prevent gastro-oesophageal reflux by constricting during inspiration (Nason et al 2012) (see also Ch. 64).

Hiatus hernia The diaphragm lends additional power to all expulsive efforts: sneezing, coughing, laughing, crying, urination, defecation and uterine fetal expulsion are all preceded by a deep inspiration. Similarly, a deep inspiration followed by closure of the glottis is a common preliminary to powerful recruitment of the trunk muscles in lifting heavy objects because the increased intra-abdominopelvic pressure pneumatically braces the vertebral column.

Repeated stress may eventually compromise the integrity of the hiatus with widening of the muscular hiatal tunnel. Concomitant laxity of the phreno-oesophageal ligament allows the gastro-oesophageal junction to migrate into the thorax; this is usually termed a sliding, or type I, hiatus hernia. Sliding hernias are typically acquired, commonly occur in the fifth decade of life, and are found in more than 50% of patients with gastro-oesophageal reflux. The latter induces tonic contraction of the longitudinal oesophageal muscle and this further exacerbates the hiatus hernia. When the stomach herniates into the thorax adjacent to the oesophagus, it is termed a para-oesophageal, or type II, hiatus hernia.

Congenital hernia Congenital diaphragmatic defects may cause abdominal organs, usually the stomach and/or the small bowel, colon, liver and spleen, to herniate through the diaphragm into the thorax; they are described on page 907.

Eventration Eventration is a focal bulge, usually in the anteromedial segment of the muscular portion of the diaphragm, that is caused by congenital thinning.

Diaphragmatic trauma



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Median arcuate ligament (Dunbar's) syndrome



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VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

ARTERIES

The profuse arterial supply of the diaphragm is derived from anastomoses between the lower five intercostal and subcostal arteries, the superior and inferior phrenic arteries, and the pericardiophrenic and musculophrenic arteries.

Intercostal and subcostal arteries

The lower five intercostal and subcostal arteries supply the costal margins of the diaphragm (Ch. 53).

Inferior phrenic arteries

The major arterial supply to the diaphragm is derived from the right and left inferior phrenic arteries (Fig. 55.4). Most commonly, the right inferior phrenic artery arises from the coeliac trunk or separately from the aorta, less commonly it arises from the right renal artery, and occasionally it arises from the left gastric artery or the hepatic artery proper. The left inferior phrenic artery usually arises from the coeliac trunk or separately from the aorta; occasionally, it may arise from the left renal artery, left gastric artery or the hepatic artery proper (Loukas et al 2008).

Each artery ascends anterolateral to the diaphragmatic crus, near the medial border of the suprarenal gland. The left passes posterior to the oesophagus and then runs anteriorly on the left side of its diaphragmatic opening. The right inferior phrenic artery passes posterior to the inferior vena cava, and then along the right side of its opening. Each phrenic artery divides into medial and lateral branches near the posterior border of the central tendon. The medial branch curves anteriorly to anastomose with its contralateral fellow anterior to the central tendon and with the musculophrenic and pericardiophrenic arteries. The lateral branch approaches the thoracic wall and anastomoses with the inferior posterior intercostal and musculophrenic arteries. The lateral branch of the right inferior phrenic artery supplies the inferior vena cava. The left sends ascending branches to the oesophagus.

Superior phrenic arteries

The superior phrenic arteries supply the superior portion of the diaphragm (Figs 55.5–55.6). The right superior phrenic artery most commonly arises from either the thoracic aorta, the proximal (pre-intercostal space) segment of the tenth intercostal artery or its distal segment. The left superior phrenic artery most commonly arises from the thoracic aorta or the proximal segment of the tenth intercostal artery; occasionally, it arises from the distal segment of the tenth intercostal artery (Loukas et al 2005).

VEINS

Phrenic veins

The superior surface of the diaphragm is drained by tributaries of the musculophrenic and pericardiophrenic veins that run alongside the corresponding arteries. The inferior surface of the diaphragm is drained by tributaries of the right and left inferior phrenic veins. The right inferior phrenic vein drains into the inferior vena cava – almost always inferior, but occasionally superior, to the diaphragm – or to the right

On imaging, it appears as a sharp transition between adjoining parts and may be mistaken for a diaphragmatic hernia. It is identified and becomes more pronounced with increased abdominal pressure and has been linked to obesity. Eventration is typically found on the left; the affected part shows diminished or paradoxical movement during fluoroscopic sniff testing while the posterior aspect of the ipsilateral dome acts normally (Nason et al 2012).

Closed or penetrating thoraco-abdominal injuries may result in rupture or laceration of the diaphragm. With closed injuries and diaphragmatic rupture, there may be subsequent herniation of the abdominal contents into the thorax. Spiral CT with planar reformation should be the primary investigation (Shanmuganathan et al 2000). Magnetic resonance imaging (MRI) is usually performed when other imaging modalities have produced equivocal findings. Early operative repair is recommended because untreated cases may develop gastrointestinal obstruction or perforation. Patients with penetrating injuries may require additional assessment by thoracoscopy (Lowdermilk and Naunheim 2000).

In up to a quarter of individuals, the median arcuate ligament lies significantly inferiorly, positioning itself anterior to the coeliac axis. This may cause compression and downward arching of the coeliac artery, leading to mesenteric angina and pancreaticoduodenal aneurysms as a consequence of resulting collateral flow. Most patients will present with postprandial pain, nausea and often vomiting, and many will avoid eating, resulting in major weight loss. During expiration, concurrent movement of the abdominal aorta and the median arcuate ligament compresses the coeliac artery; the position, angle and level of the coeliac artery are also major aetiological factors. Diagnosis is based mainly on exclusion of other causes, e.g. compression of the vessel as a result of fibrotic or neoplastic changes in the sympathetic plexus and coeliac ganglion, as well as atherosclerotic changes in the vessel itself. Definitive diagnosis can only be made with multiple imaging modalities.



Fig. 55.4 A posterior cadaveric dissection showing the origin and course of the right (RIPA) and left (LIPA) inferior phrenic arteries. The right inferior phrenic artery arises from the superior mesenteric artery (SMA), and the left inferior phrenic artery arises from the coeliac trunk. (With permission from Loukas M, Hullett J, Wagner T. Clinical anatomy of the inferior phrenic artery. Clin Anat. 2005;18(5):357–65.)

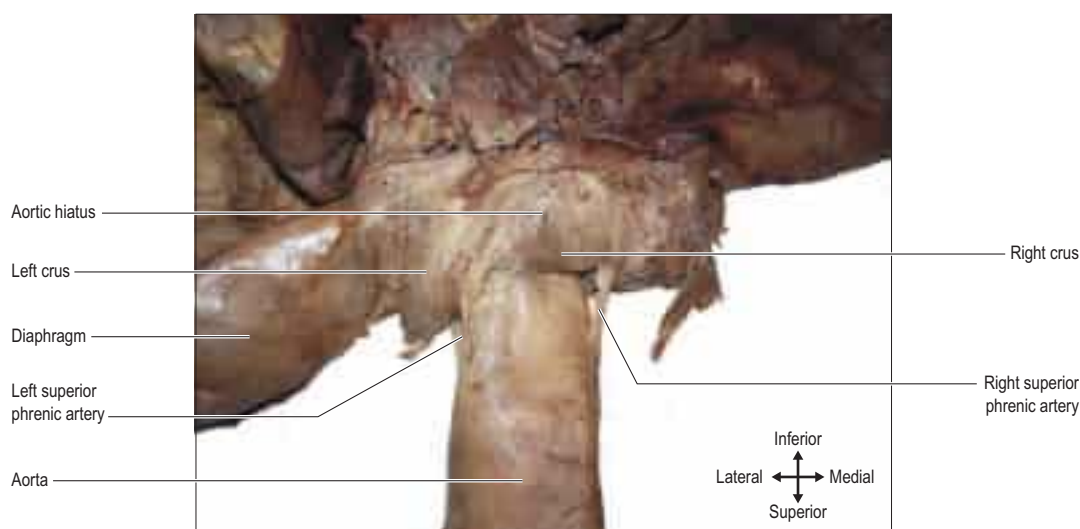


Fig. 55.5 A cadaveric dissection showing the origin of the right and left superior phrenic arteries from the thoracic aorta and supplying portions of the right and left diaphragmatic crura. (With permission. Loukas M, Louis RG Jr, Wartmann CT, Tubbs RS, Esmaeili E, Bagenholm AC, Merbs W, Curry B, Jordan R. Superior phrenic artery: an anatomic study. Surg Radiol Anat. 2007;29(1):97–102.)

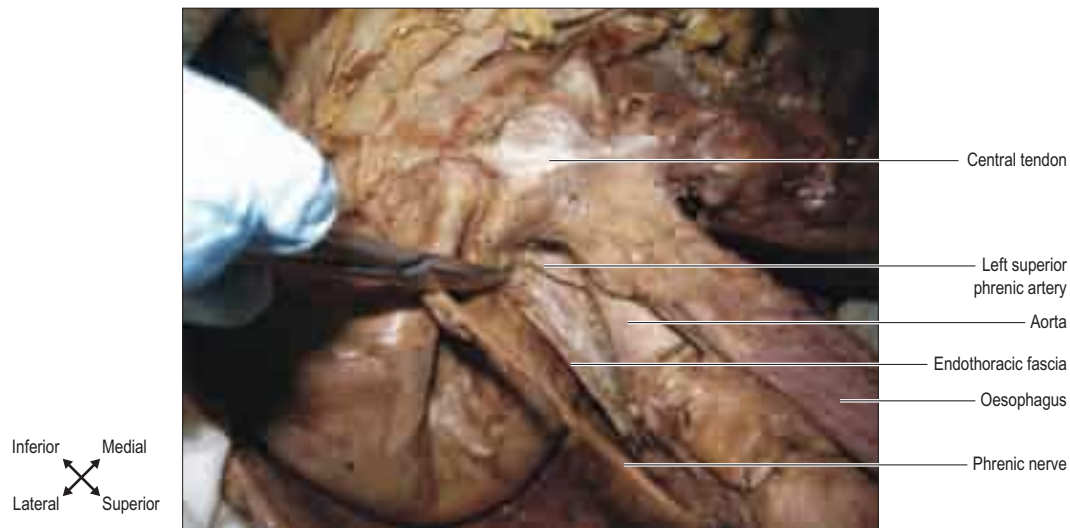


Fig. 55.6 A lateral view of the left superior phrenic artery arising from the aorta. (With permission. Loukas M, Louis RG Jr, Wartmann CT, Tubbs RS, Esmaeili E, Bagenholm AC, Merbs W, Curry B, Jordan R. Superior phrenic artery: an anatomic study. *Surg Radiol Anat.* 2007;29(1):97-102.)

hepatic vein. The left inferior phrenic vein drains into the inferior vena cava – commonly inferior, or rarely superior, to the diaphragm. It may also drain into the left suprarenal vein, the left renal vein, the left hepatic vein or, rarely, the left suprarenal renal vein and the inferior vena cava (Loukas et al 2005).

LYMPHATIC DRAINAGE

The diaphragm is drained by anterior, middle and posterior groups of lymph nodes that lie on its superior surface; they also drain the superior portion of the liver, the gastro-oesophageal junction and the abdominal surface of the diaphragm. In addition, lymph drains anterosuperiorly to parasternal and anterior mediastinal nodes, and posterosuperiorly to posterior mediastinal and brachiocephalic nodes.

Hepatocellular carcinoma



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INNERVATION

The diaphragm receives its motor supply via the phrenic nerves (Fig. 55.7). Sensory fibres are distributed to the peripheral part of the muscle by the lower six or seven intercostal nerves. The fibres of the right crus of the diaphragm divide to the right and left of the oesophagus and are innervated by both right and left phrenic nerves. There is some evidence that the crural fibres contract slightly before the costal part, and this may be functionally significant for non-respiratory tasks.

Phrenic nerve

The phrenic is a mixed nerve that provides the sole motor supply to its hemidiaphragm. Derived mostly from the fourth cervical ramus, it also receives contributions from the third and fifth cervical ventral rami (see Figs 29.17, 29.18). The course of the cervical part of the phrenic nerve is described on page 464. Within the thorax, the phrenic nerve descends anterior to the pulmonary hilum between the fibrous pericardium and mediastinal pleura, accompanied by the pericardiophrenic vessels. In its thoracic course, each phrenic nerve supplies sensory branches to the mediastinal pleura, fibrous pericardium and parietal serous pericardium. The right and left phrenic nerves differ in their intrathoracic relationships (Rajanna 1947).

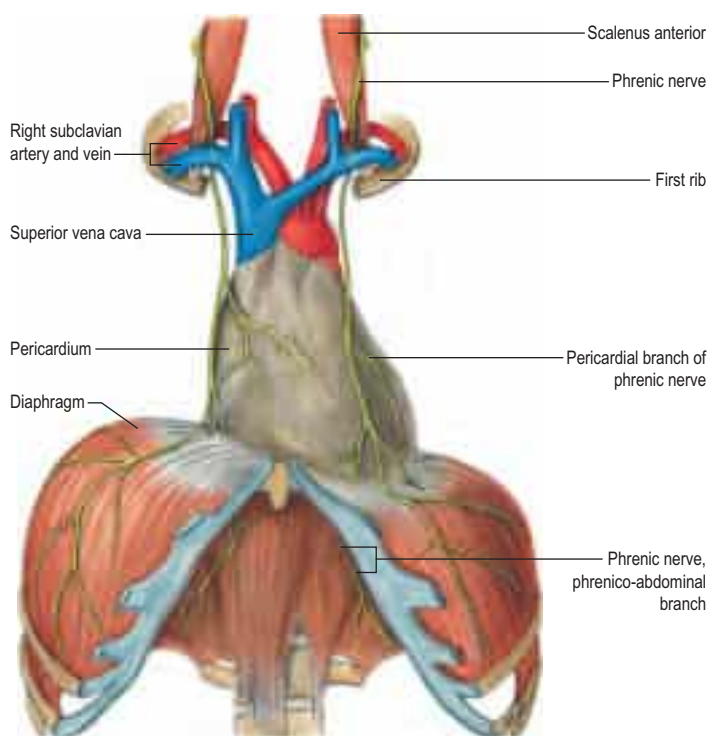


Fig. 55.7 The distribution of the right and left phrenic nerves. (With permission from Waschke J, Paulsen F (eds), *Sobotta Atlas of Human Anatomy*, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

Right phrenic nerve

The right phrenic nerve is shorter and more vertical than the left. Separated at the root of the neck from the second part of the right subclavian artery by scalenus anterior, the nerve then descends lateral to the right brachiocephalic vein, the superior vena cava and the fibrous pericardium covering the right surface of the right atrium and inferior vena cava, and divides just superior to or at the level of the diaphragm.

Left phrenic nerve

At the root of the neck, the left phrenic nerve commonly leaves the medial edge of scalenus anterior and passes anterior to the first part of the left subclavian artery and posterior to the thoracic duct. However, sometimes the right and left nerves are symmetrical in their cervical course, in which case the left phrenic nerve may cross anterior to the second part of the left subclavian artery, from which it is separated by scalenus anterior, at the level of the thoracic inlet. Thereafter, the left phrenic nerve crosses anterior to the left internal thoracic artery and descends across the medial aspect of the apex of the left lung and its pleura to the first part of the subclavian artery, which it crosses obliquely to reach a groove between the left common carotid and subclavian arteries. It passes anteromedially, superficial to the left vagus nerve just superior to the aortic arch and posterior to the left brachiocephalic vein, then descends superficial to the aortic arch and the left superior intercostal vein, anterior to the left pulmonary hilum, to lie between the fibrous pericardium covering the surface of the left ventricle and the mediastinal pleura (see Fig. 57.3).

Diaphragmatic relationships The right phrenic nerve passes through the central tendon of the diaphragm, either by the caval hiatus or just lateral to it. The left phrenic nerve passes through the muscular part of the diaphragm anterior to the central tendon, just lateral to the left cardiac surface and more anterior than the right phrenic nerve (see Fig. 55.7). At the diaphragm or slightly superior to it, each phrenic nerve supplies fine branches to the parietal pleura above the central diaphragm, and the parietal peritoneum below. Typically, the trunk of each nerve then divides into three branches as it passes through the diaphragm. With some variation, these constitute an anterior (sternal) branch, which runs anteromedially towards the sternum and connects with its fellow; an anterolateral branch, which runs to the lateral leaf of the central tendon; and a short posterior branch. The latter divides into a posterolateral ramus that courses posterior to the lateral leaf and a posterior ramus that supplies the crural fibres. Posterolateral and posterior crural rami may arise as separate branches from the phrenic nerve.

These main branches may be submerged in diaphragmatic muscle or lie inferior to it. They supply motor fibres to the muscle and sensory fibres to the peritoneum and pleura related to the central part of the diaphragm. They also relay proprioceptive fibres from the musculature. The right phrenic nerve supplies the part of the right crus that lies to the right of the oesophagus, and the left phrenic nerve supplies the left crus and the part of the right crus that lies on the left of the oesophagus.

Phrenic ganglia The right phrenic nerve continues its course inferior to the diaphragm and frequently connects to the right phrenic ganglion. An additional, smaller, right accessory phrenic ganglion is sometimes located proximally along the phrenic nerve, closer to its caval foraminal exit. The right phrenic ganglion exhibits plexiform communications with the coeliac and aorticorenal ganglia and with the right suprarenal gland (Fig. 55.8). The left phrenic nerve usually continues its course inferior to the diaphragm. A left phrenic ganglion is less common than its right counterpart; when present, it exhibits plexiform communications to several autonomic ganglia. Both phrenic nerves and ganglia are closely related to the diaphragmatic crura; sutures for crural approximation in repair of hiatal hernia must be placed superior to the ganglia in order to avoid iatrogenic injuries to the autonomic supply of the diaphragm and abdomen (Loukas et al 2015b).

Lesions of the phrenic nerve An appreciation of the location of the main branches of the phrenic nerves is important to avoid iatrogenic surgical damage. Radial incisions in the diaphragm from the costal margin to the oesophageal hiatus lead to diaphragmatic paralysis, whereas thoraco-abdominal incisions in a circumferential manner in the periphery of the diaphragm do not involve any significant branches of the phrenic nerves and preserve diaphragmatic function. Similarly, incisions of the central tendon are safe.

Division of the phrenic nerve in the neck completely paralyses the corresponding half of the diaphragm, which atrophies. If an accessory phrenic nerve exists, section or crushing of the main nerve as it lies on

Successful control of hepatocellular carcinoma often requires chemoembolization of the hepatic artery, as well as any extrahepatic collaterals; the right inferior phrenic artery is often found to be associated with hepatocellular carcinoma as a major collateral pathway. Transcatheter oily chemoembolization regresses the tumour with increased survival rates. The inferior phrenic vein is one of the major sources of collateral venous drainage in hepatocellular carcinoma (and in retroperitoneal malignant disease). As well as conventional angiography, the use of 64+ slice computed tomography (CT) scanners enables clear visualization of the right and left inferior phrenic arteries, making CT a reliable method of diagnosing thrombosis and aneurysm of these vessels (Loukas et al 2008).

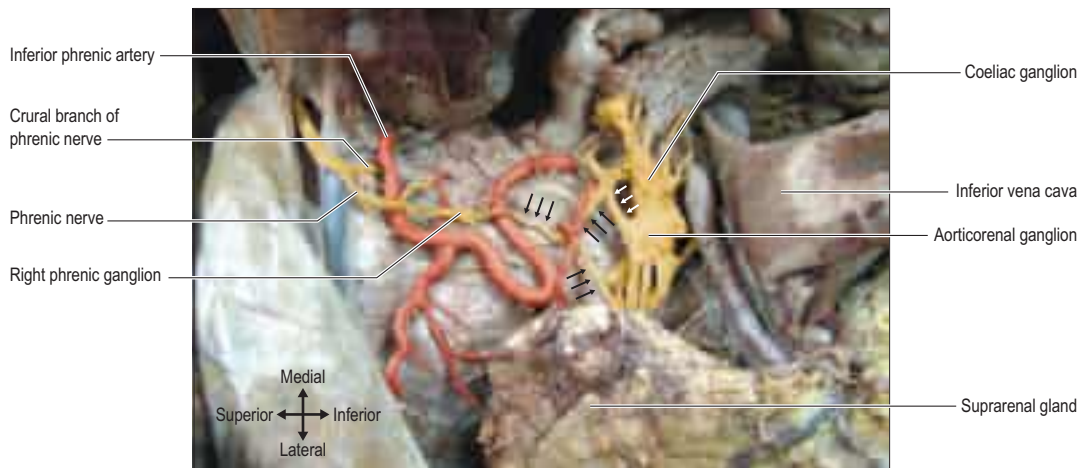


Fig. 55.8 A cadaveric dissection showing a right phrenic ganglion close to the right inferior phrenic artery and right diaphragmatic crus with plexiform communications to the coeliac ganglion, aorticorenal ganglion and suprarenal gland (arrows). (With permission from Loukas M, Joseph S, Louis RG Jr., Wartmann CT, Tubbs RS. The surgical anatomy of the subdiaphragmatic phrenic nerve and phrenic ganglia. *Clinical Anatomy*, in press, 2015b.)

scalenus anterior will not produce complete paralysis. The phrenic nerve may be involved in traumatic lesions of the upper brachial plexus. Historically, it was deliberately injured in order to collapse, and hence rest, the lung in patients with pulmonary tuberculosis. Cardiac surgery is one of the most common iatrogenic causes of phrenic nerve injury, especially as a result of the instillation of saline slush for myocardial preservation. Surgery for tumours of the lung or mediastinum may also cause damage, as well as infections such as poliomyelitis, obstetric injury during traumatic childbirth, and radiation therapy (Nason et al 2012). Respiratory muscle weakness of rapid onset is a feature of Guillain-Barré syndrome.

Phrenic nerve damage leads to paradoxical movement of the diaphragm that is best observed fluoroscopically, with the patient first in the upright position (diaphragm unloaded), and then supine with a small weight on the abdomen (diaphragm loaded). Diaphragmatic paralysis can also be assessed by ultrasound examination with a sniff manoeuvre. The 'gold standard' for assessment of the phrenic nerve/diaphragm unit is electrical or magnetic stimulation of the phrenic nerve with recording of the compound muscle action potential and/or the transdiaphragmatic pressure. Electrical stimulation of the diaphragm, by 'pacing' of one or both phrenic nerves, has been used with some success in infants with central alveolar hypoventilatory syndrome ('Ondine's curse') and in patients with high cervical lesions of the spinal cord, in whom the diaphragm is paralysed but the lower motor neurones are intact. Electrodes are placed adjacent to the nerves, sometimes in the neck but more usually in the chest, and a ventilatory rhythm is established by trains of stimuli delivered by an implanted device. As this is a non-physiological way of recruiting the muscle, the fibres must be 'conditioned' during the initial period of stimulation, so that they acquire the necessary resistance to fatigue.

Referred pain Diaphragmatic pain is frequently felt at the tip of the shoulder, reflecting common nerve root origins in the neck. It usually occurs when there is inflammation of the diaphragmatic pleura, e.g. in basal pneumonia, pleural effusions or malignant disease, or with irritation of the peritoneal pleura, as with subphrenic abscess, where involuntary diaphragmatic contraction (hiccups) may also occur.

Accessory phrenic nerve

The accessory phrenic nerve is described on [page 464](#).



ANATOMY OF BREATHING

Breathing is a highly coordinated abdominal and thoracic process. The diaphragm is the major muscle of inspiration, responsible for approximately two-thirds of quiet breathing in healthy humans. The external intercostal muscles are most active in inspiration, and the less strong internal intercostals in expiration. Increasing the vertical, transverse and anteroposterior dimensions of the chest increases the volume of the pleural space, and the resulting decrease in intrapleural pressure draws air into the lungs. During expiration, the diaphragm relaxes and moves superiorly. Air is expelled from the lungs and the elastic recoil of the lung creates a subatmospheric pressure that returns the lateral and anteroposterior dimensions of the thorax to normal (De Troyer and Estenne 1988, Celli 1998). During inspiration, the most inferior ribs are fixed and contraction of the diaphragm draws the central tendon inferiorly, scarcely altering diaphragmatic curvature. The cupulae move inferiorly and a little anteriorly, almost parallel to their original positions. The associated inferior displacement of the abdominal viscera is permitted by the extensibility of the abdominal wall, but the limit of this extensibility is soon reached. The central tendon, its motion arrested by the abdominal viscera, then becomes a fixed point from which the fibres of the diaphragm continue to contract. This causes the second to tenth ribs to be elevated and the inferior portions of the ribs are turned outwards as a result of direct transmission of pressure through the zone of apposition (Fig. 55.9). The medial aspect of the rib is elevated and this increases the transverse dimension of the chest in the same manner as a bucket handle swinging outwards; this effect is most evident in the inferior seventh to tenth ribs. Movements at the costovertebral joints cause elevation of the anterior ends of the ribs that push the body of the sternum and the upper ribs anteriorly in a 'pump handle' movement; this effect is most evident in the superior second to sixth ribs, and increases the anteroposterior dimension of the thorax. The right cupula of the diaphragm, which lies on the liver, has a greater resistance to overcome than the left, which lies over the stomach, and so the right crus and the fibres of the right side are more substantial

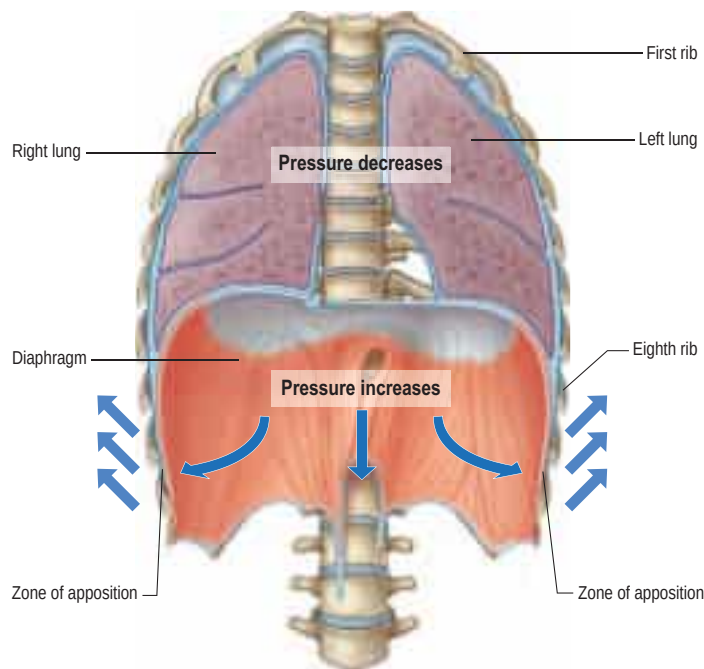


Fig. 55.9 Inspiratory movements: pressure changes during inspiration. (Adapted from Drake RL, Vogl AW, Mitchell A (eds), *Gray's Anatomy for Students*, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

than those of the left. The balance between descent of the diaphragm, protrusion of the abdominal wall ('abdominal breathing'), and elevation of the ribs ('thoracic breathing') varies in different individuals and with the phase and depth of ventilation, the degree of distension of the stomach and intestines, and the size of the liver. The thoracic element is usually more marked in females, but increases in both sexes during deep inspiration.

Radiographs show that the height of the diaphragm within the thorax also varies considerably with posture. It is highest when the body is supine, when it performs the greatest ventilatory excursions with normal breathing. When the body is erect, the diaphragm is lower and its ventilatory movements become smaller. The diaphragmatic profile is still lower in the sitting posture and ventilatory excursions are smallest under these conditions. When the body is horizontal and on one side, the two halves of the diaphragm do not behave in the same way. The uppermost half sinks to a lower level than that seen when sitting, and moves little with ventilation. The lower half rises higher in the thorax than it does even in the supine position, and its ventilatory excursions are considerably greater. Changes in the level of the diaphragm with alterations in posture explain why patients with severe dyspnoea are most comfortable, and least short of breath, when sitting up.

Diaphragmatic excursion is typically 1.5 cm in quiet breathing. During deep ventilation, the maximum movement ranges from 6 to 10 cm. After a forced inspiration, e.g. when breathing is partially obstructed, the right cupula of the diaphragm may descend to about the level of the eleventh thoracic vertebra, while the left cupula may reach the level of the body of the twelfth thoracic vertebra. After forced expiration, the right cupula of the diaphragm is level anteriorly with the fourth costal cartilage, laterally with the fifth, sixth and seventh ribs, and posteriorly with the eighth rib; the left cupula is a little lower.

The role of the abdomen in breathing is often underestimated. If, for example, the anterolateral wall were made of steel and linked the pelvic rim rigidly to the costal margins, inspiration would be impossible; the diaphragm could not descend (because the abdominal contents are incompressible), and the ribs could not rise (because the links to the pelvis would be inextensible). During normal breathing, the abdomen relaxes as the diaphragm contracts. It is possible to oppose this motion by tensing the abdomen, as in the 'beach posture' adopted to exaggerate the size of the chest. In this case, the abdominal contents fix the central tendon of the diaphragm, so that it raises the ribcage as it contracts, but it is a condition of that manoeuvre that the gap between the ribs and the pelvic rim widens.

The ventilatory muscles must also work during sleep, when the pharyngeal muscles relax and upper airway resistance increases. It is now appreciated that in some people, particularly the obese, this relaxation can lead to periodic apnoea and marked hypoxia during sleep, implying that the pharyngeal muscles play an important ventilatory role

in waking life. It is also clear that although ventilatory muscles rarely tire in normal life, they do fatigue when placed under abnormal loads, e.g. in chronic obstructive pulmonary disease.

The different pulmonary regions do not all move equally in ventilation. In quiet ventilation, the juxtahilar part of the lung scarcely moves and the middle region moves only slightly. The superficial parts of the lung expand the most, and the mediastinal surface, posterior border and apex move less because they are related to less movable

structures. The diaphragmatic and costomediastinal regions expand most of all. Most of the volumetric change during ventilation occurs in the alveoli.

The primary role of the intercostal muscles is to stiffen the chest wall, preventing paradoxical motion during inspiratory diaphragmatic descent. This becomes most obvious immediately after high spinal injury, when there is flaccid paralysis of the entire trunk and only the diaphragm remains functional.



Bonus e-book images

Fig. 55.4 A cadaveric dissection showing the origin and course of the right and left inferior phrenic arteries.

Fig. 55.5 A cadaveric dissection showing the origin of the right and left superior

phrenic arteries from the thoracic aorta and supplying portions of the right and left diaphragmatic crura.

Fig. 55.6 A lateral view of the left superior phrenic artery arising from the aorta.

Fig. 55.8 A cadaveric dissection showing a right phrenic ganglion close to the right inferior phrenic artery and right diaphragmatic crus with plexiform communications to the coeliac ganglion, aorticorenal ganglion and suprarenal gland.

In a healthy adult with a vital capacity of 4.5 litres, some 3 litres is accounted for by diaphragmatic excursion. Immediately after high spinal injury, the vital capacity decreases to about 300 ml, even though the diaphragm is moving maximally, because some 2.7 litres is lost by paradoxical incursion of the flaccid chest wall as the diaphragm descends. Usually after several weeks, the paralysis becomes spastic, stiffening the chest wall, and the vital capacity increases towards its phrenic limit of about 3 litres. In the same way, high spinal injury reveals the role of the abdomen in inspiration and expiration. The abdominal musculature plays a major role in active expiration in humans. During the flaccid stage of high spinal paralysis, the only mechanisms available for returning the relaxed diaphragm into the thorax on expiration are passive recoil of the lungs and chest wall, and the weight of the abdominal viscera. The latter is the most important and operates only when patients are lying down. If they are sat up or raised upright, they are unable to exhale; trussing the abdomen with an elastic binder can be helpful. Conversely, when paralysis becomes spastic, the stiff abdominal wall opposes inspiration.

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